12

CHAPTER

Transient Currents (Varying Currents)

Learning Objectives

After reading this chapter, you should be able to:

- Derive Helmholtz's equations for the growth and decay of electric current in a circuit containing a resistance and an inductance.
- Discuss the phenomenon of charging of a capacitor through a series resistance. Define time constant of an *R–C* circuit. Explain the significance of time constant.
- Discuss the discharge of a capacitor in a purely resistive circuit and explain how the result may be used for the measurement of a high resistance of the order of megohm.
- Explain why a spark appears when a high inductance circuit is suddenly opened but not at make.
- Deduce the condition necessary for maintaining an oscillatory discharge of a capacitor in a circuit containing a resistance and an inductance.

Growth and decay of currents in circuits with resistance, capacitance and inductance (Helmholtz's equations).

12.1 GROWTH OF CURRENT IN A CIRCUIT CONTAINING A RESISTANCE AND INDUCTANCE

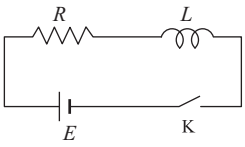


FIG. 12.1

Consider a circuit having an inductance L and a resistance R connected in series to a cell of steady emf E (Fig. 12.1). When the key K is pressed, there is a gradual growth of current in the circuit from zero to maximum value I_0. Let I be the instantaneous current at any instant.

Then, the induced back emf $\varepsilon = -L\dfrac{dI}{dt}$

$$\therefore \quad E = RI + L\frac{dI}{dt} \qquad \ldots(1)$$

When the current reaches the maximum value I_0, the back emf,

$$L\frac{dI}{dt} = 0$$

Hence $$E = RI_0 \qquad \ldots(2)$$

Substituting this value for E in Eq. (1),

$$RI_0 = RI + L\frac{dI}{dt} \text{ or } R(I_0 - I) = L\frac{dI}{dt}$$

or $$\frac{dI}{I_0 - I} = \frac{R}{L}dt$$

Integrating, $$-\log (I_0 - I) = \frac{R}{L}t + C \qquad \ldots(3)$$

where C is the constant of integration.

When $t = 0$, $I = 0$, $\quad \therefore -\log_e I_0 = C$

Substituting this value of C in Eq. (3),

$$-\log (I_0 - I) = \frac{R}{L}t - \log_e I_0 \quad \text{or} \quad \log_e (I_0 - I) - \log_e I_0 = -\frac{R}{L}t$$

$$\log_e \frac{(I_0 - I)}{I_0} = -\frac{R}{L}t$$

$$\frac{I_0 - I}{I_0} = e^{-(R/L)t} \text{ or } 1 - \frac{I}{I_0} = e^{-(R/L)t}$$

$$\therefore \quad I = I_0 (1 - e^{-(R/L)t}) \qquad \ldots(4)$$

Eq. (4) gives the value of the instantaneous current in the LR circuit. The quantity (L/R) is called the *time constant* of the circuit.

$$\text{If } \frac{L}{R} = t,\ I = I_0(1 - e^{-1}) = I_0\left(1 - \frac{1}{e}\right) = 0.632 I_0$$

Thus, *the time constant L/R of a L-R circuit is the time taken by the current to grow from zero to 0.632 times the steady maximum value of current in the circuit.*

Similarly, when $t = 2L/R$, $3L/R$ …, the value of current will be 0.8647, 0.9502, … of the final maximum current.

When $t = 0$, $I = 0$ and when $t \to \infty$, $I = I_0$.

Greater the value of L/R, longer is the time taken by the current I to reach its maximum value (Fig. 12.2).

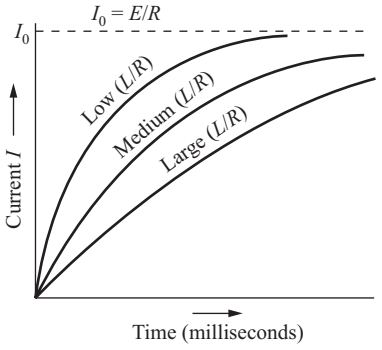


FIG. 12.2

12.2 DECAY OF CURRENT IN A CIRCUIT CONTAINING L AND R

When the circuit is broken, an induced emf, equal to $-L\dfrac{dI}{dt}$ is again produced in the inductance L and it slows down the rate of decay of the current. The current in the circuit decays from maximum value I_0 to zero. During the decay, let I be the current at time t. In this case $E = 0$. The emf equation for the decay of current is

$$0 = RI + L\frac{dI}{dt} \qquad \ldots(1)$$

$$\therefore \quad \frac{dI}{I} = -\frac{R}{L}dt$$

Integrating, $\log_e I = -\dfrac{R}{L}t + C$ where C is a constant

When $t = 0$, $I = I_0$ $\quad \therefore \quad \log_e I_0 = C$

$$\log_e I = -\frac{R}{L}t + \log_e I_0, \quad \text{or} \quad \log_e \frac{I}{I_0} = -\frac{R}{L}t$$

$$\frac{I}{I_0} = e^{-(R/L)t}$$

$$\therefore \quad I = I_0 e^{-(R/L)t} \qquad \ldots(2)$$

Eq. (2) represents the current at any instant t during decay. A graph between current and time is shown in Fig. 12.3.

When $t = L/R$, $I = I_0 e^{-1} = \dfrac{1}{e} I_0 = 0.365\, I_0$

$t = 2L/R$, $I = I_0 e^{-2} = 0.1035\, I_0$

$t = 3L/R$, $I = I_0 e^{-3} = 0.05\, I_0$

Therefore, *the time constant L/R of a R–L circuit may also be defined as the time in which the current in the circuit falls to 1/e of its maximum value* when external source of e.m.f. is removed.

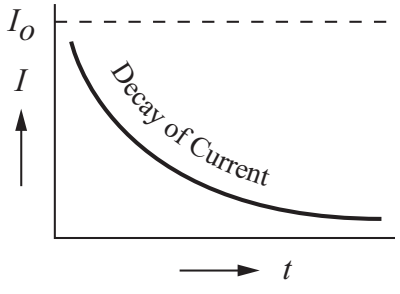


FIG. 12.3

The rate of decay of current is

$$\frac{dI}{dt} = -\frac{R}{L} I_0 e^{-(R/L)t} = -\frac{R}{L} I$$

Thus it is clear that greater the ratio *R/L, or smaller the time constant L/R, the more rapidly does the current die away* (Fig. 12.4).

Fig. 12.5 shows that the growth and decay curves are complementary.

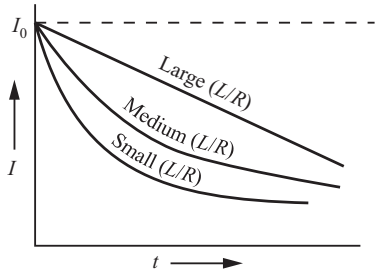


FIG. 12.4

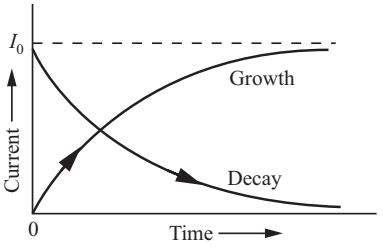


FIG. 12.5

Example 1. *An e.m.f. 10 volts is applied to a circuit having a resistance of 10 ohms and an inductance of 0.5 henry. Find the time required by the current to attain 63.2% of its final value. What is the time constant of the circuit?*

Solution. $I = I_0(1 - e^{-Rt/L})$

Given: $\frac{I}{I_0} = \frac{63.2}{100}; \frac{R}{L} = \frac{10}{0.5} = 20$

$$\frac{63.2}{100} = 1 - e^{-20t}$$

$$e^{-20t} = 1 - 0.632 = 0.368$$

$$e^{20t} = \frac{1}{0.368} = 2.717$$

$$20t = \log_e 2.717$$

$$t = \frac{1}{20} \times 2.3026 \times \log_{10} 2.717$$

$$= \frac{2.3026 \times 0.4341}{20} = \mathbf{0.05\ sec.}$$

The time constant of the circuit is

$$\frac{L}{R} = \frac{0.5}{10} = \frac{1}{20} \text{ sec.}$$

Example 2. *An inductance of 500 mH and a resistance of 5 ohms are connected in series with an e.m.f of 10 volts. Find the final current. If now the cell is removed and the two terminals are connected together, find the current after (i) 0.05 sec. and (ii) 0.2 sec.*

Solution. Final current $I_0 = \frac{E}{R} = \frac{10}{5} = 2A$

During discharge, $I = I_0 e^{-Rt/L}$

Now $\frac{R}{L} = \frac{5}{500 \times 10^{-3}} = 10$

(*i*) When $t = 0.05$ *sec*., $I = 2e^{-10 \times 0.05} = 1.213\,A$

(*ii*) When $t = 0.2$ *sec*., $I = 2e^{-10 \times 0.2} = 0.271\,A$

12.3 GROWTH AND DECAY OF CHARGE IN A CIRCUIT CONTAINING RESISTANCE AND CAPACITOR (CHARGING AND DISCHARGING OF A CAPACITOR THROUGH A RESISTOR)

(*a*) Growth of Charge. A capacitor C and a resistance R are connected to a cell of emf E through a Morse key K (Fig. 12.6). When the key is pressed, a momentary current I flows through R. At any instant t, let Q be the charge on the capacitor of capacitance C.

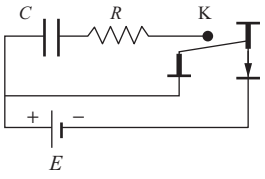


FIG. 12.6

P.D. across capacitor = Q/C

P.D. across resistor = RI

The emf equation of the circuit is

$$E = (Q/C) + RI \qquad \ldots(1)$$

$$E = (Q/C) + R\,(dQ/dt) \qquad \because \quad I = dQ/dt$$

The capacitor continues getting charged till it attains the maximum charge Q_0. At that instant $I = dQ/dt = 0$. The P.D. across the capacitor is $E = Q_0/C$.

i.e., when, $$Q = Q_0, \frac{dQ}{dt} = 0 \text{ and } E = \frac{Q_0}{C}$$

$\therefore$ $$\frac{Q_0}{C} = \frac{Q}{C} + R\frac{dQ}{dt}$$

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$

$$\left(\frac{dQ}{Q_0 - Q}\right) = \frac{dt}{CR} \qquad \ldots(2)$$

Integrating, $-\log_e (Q_0 - Q) = \frac{t}{CR} + K$

where K is a constant.

When $t = 0, Q = 0 \quad \therefore \quad -\log_e Q_0 = K$

$$-\log_e (Q_0 - Q) = \frac{t}{CR} - \log_e Q_0$$

$$\log_e (Q_0 - Q) = -\frac{t}{CR} + \log_e Q_0$$

$$\log_e (Q_0 - Q) - \log_e Q_0 = -\frac{t}{CR}$$

$$\log_e \left(\frac{Q_0 - Q}{Q_0}\right) = -\frac{t}{CR}$$

$$\frac{Q_0 - Q}{Q_0} = e^{-\frac{t}{CR}} \text{ or } 1 - \frac{Q}{Q_0} = e^{-\frac{t}{CR}}$$

$$\therefore \quad Q = Q_0\left(1 - e^{-\frac{t}{CR}}\right) \qquad ...(3)$$

The term CR is called time constant of the circuit.

At the end of time $t = CR, Q = Q_0\,(1 - e^{-1}) = 0.632\,Q_0$

Thus, *the time constant may be defined as the time taken by the capacitor to get charged to 0.632 times its maximum value.*

The growth of charge is shown in Fig. 12.7.

The rate of growth of charge is

$$\frac{dQ}{dt} = \frac{Q_0}{CR}\,e^{-\frac{t}{CR}} = \frac{I}{CR}\,(Q_0 - Q)$$

Thus it is seen that smaller the product CR, the more rapidly does the charge grow on the capacitor.

The rate of growth of the charge is rapid in the beginning and it becomes less and less as the charge approaches nearer and nearer the steady value.

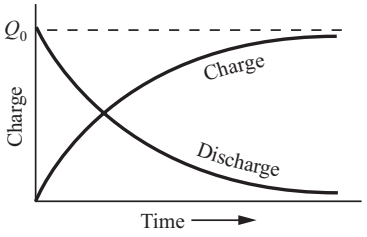


FIG. 12.7

(*b*) Decay of charge (Discharging of a Capacitor through Resistance)

Let the capacitor having charge Q_0 be now discharged by releasing the Morse key K (Fig. 12.6). The charge flows out of the capacitor and this constitutes a current. In this case $E = 0$.

$$R\,\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad \ldots(1)$$

or $$\frac{dQ}{Q} = -\frac{1}{CR}\,dt$$

Integrating, $\log_e Q = -\dfrac{t}{CR} + K,$ where K is a constant

When $t = 0, Q = Q_0 \therefore \log_e Q_0 = K$

$$\log_e Q = -\frac{t}{CR} + \log_e Q_0$$

or $$\log_e \frac{Q}{Q_0} = -\frac{t}{CR} \quad \text{or} \quad \frac{Q}{Q_0} = e^{-t/CR}$$

$$\therefore \quad Q = Q_0 e^{-t/CR} \qquad \ldots(2)$$

This shows that the charge in the capacitor decays exponentially and becomes zero after infinite interval of time (Fig. 12.7).

The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}\,e^{-t/CR} = -\frac{Q}{CR} \qquad \ldots(3)$$

Thus, smaller the time-constant CR, the quicker is the discharge of the capacitor.

In Eq. (2), if we put $t = CR$, then $Q = Q_0 e^{-1} = 0.368\,Q_0$.

Hence *time constant may also be defined as the time taken by the current to fall from maximum to 0.368 of its maximum value.*

Example 1. *A capacitor is charged by DC supply through a resistance of 2 megaohms. If it takes 0.5 seconds for the charge to reach three quarters of its final value, what is the capacitance of the capacitor?*

Solution. Here, $R = 2 \times 10^6\ \Omega$, $t = 0.5$s, $Q/Q_0 = 3/4$.

$$Q = Q_0\,(1 - e^{-t/CR}) \quad \text{or} \quad \frac{Q}{Q_0} = (1 - e^{-t/CR})$$

or $$\frac{3}{4} = \left(1 - e^{-\frac{0.5}{C\times(2\times10^6)}}\right)$$

or $$e^{\frac{0.5}{C\times(2\times10^6)}} = 4 \quad \text{or} \quad \frac{0.5}{C\times(2\times10^6)} = \log_e 4$$

or $$C = \frac{0.5}{(2.3026 \log_{10} 4)\,(2\times10^6)} = 0.18\times10^{-6}\ \text{F} = 0.18\ \mu\text{F}.$$

Example 2. *A capacitor of capacitance 0.1 μF is first charged and then discharged through a resistance of 10 megaohm. Find the time, the potential will take to fall to half its original value.*

Solution. $Q = Q_0\ e^{-t/CR}$ or $\frac{Q}{Q_0} = e^{-t/CR}$ or $\ln\left(\frac{Q_0}{Q}\right) = \frac{t}{CR}$

$\therefore$ $$t = CR \ln\left(\frac{Q_0}{Q}\right)$$

$Q = CV$ and $Q_0 = CV_0$. $\therefore$ $\frac{Q_0}{Q} = \frac{V_0}{V}$

$\therefore$ $$t = CR \ln\left(\frac{V_0}{V}\right)$$

Here, $C = 10^{-7}\ F$; $R = 10^7\ \Omega$, $V_0/V = 2$.

$\therefore$ $t = 10^{-7} \times 10^7 \times \ln 2 = 0.6931$ s.

Example 3. A *resistance R and a 2μF capacitor in series are connected to a 200 volt direct supply. Across the capacitor is a neon lamp that strikes at 120 volts. Calculate the value of R to make the lamp strike 5 seconds after switch has been closed.*

Solution. The resistance R must be such that the p.d. across the capacitor should rise to 120 volt in 5 seconds after the switch is closed. The lamp would then strike,

The equation of charging is

$$Q = Q_0(1 - e^{-t/RC})$$

or $$CV = CV_0(1 - e^{-t/RC})$$

or $$V = V_0\,(1 - e^{-t/RC})$$

Here, $V = 120$ volt, $V_0 = 200$ volt, $t = 5$ sec and

$C = 2 \times 10^{-6}$ farad,

$\therefore$ $$120 = 200\,(1 - e^{-5/2\times10^{-6}R})$$

or $$e^{5/2\times10^{-6}R} = \frac{5}{2}$$

$$\frac{5}{2\times10^{-6}R} = \log_e 5 - \log_e 2 = 1.6094 - 0.6931 = 0.9163$$

or $$R = \frac{5}{2\times10^{-6}\times0.9163} = 2.73\times10^6\ \text{ohm} = \mathbf{2.73\ megaohm}$$

12.4 MEASUREMENT OF HIGH RESISTANCE BY LEAKAGE

Principle. When a capacitor of capacitance C and initial charge Q_0 is allowed to discharge through a resistance R for a time t, the charge remaining on the capacitor is given by

$$Q = Q_0 e^{-t/CR}$$

$$\frac{Q_0}{Q} = e^{t/CR}$$

$$\log_e \frac{Q_0}{Q} = \frac{t}{CR}$$

$$\therefore \quad R = \frac{t}{C \log_e (Q_0/Q)} = \frac{t}{2.3026\, C \log_{10} (Q_0/Q)}$$

If R is high, CR will be high and the rate of discharge of capacitor will be very slow. Thus if we determine Q_0/Q from experiment, then R can be calculated.

Circuit diagram. Connections are made as shown in Fig. 12.8. C is a capacitor of known capacitance, R is the high resistance to be measured, B.G. is a ballistic galvanometer, E is a cell, and K_1, K_2, K_3 are tap keys.

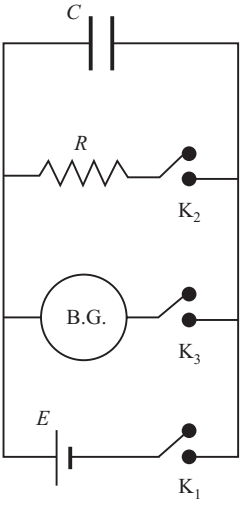


FIG. 12.8

Procedure (*i*) Keeping K_2 and K_3 open, the capacitor is charged by depressing the key K_1. K_1 is then opened and at once K_3 is closed. The capacitor discharges through the galvanometer which records a throw θ_0. The throw θ_0 is proportional to Q_0 *i.e.*, $Q_0 \propto \theta_0$.

(*ii*) The capacitor is again charged to the maximum value keeping K_2 and K_3 open and closing K_1. K_1 is then opened and K_2 is closed for a *known time t*. Some of the charge leaks through R**.** K_2 is opened and at once K_3 is closed. The charge Q remaining on the capacitor then discharges through the galvanometer. The resulting throw θ is noted. Then $Q \propto \theta$.

Now, $$\frac{Q_0}{Q} = \frac{\theta_0}{\theta}$$

$$\therefore \quad R = \frac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$$

A series of values of t and θ are obtained.

Graph. A graph is plotted between t and $\log_{10} (\theta_0/\theta)$ which is a straight line (Fig. 12.9). Its slope gives the mean value of $\frac{t}{\log_{10} (\theta_0/\theta)}$. As C is known, the value of R can be calculated.

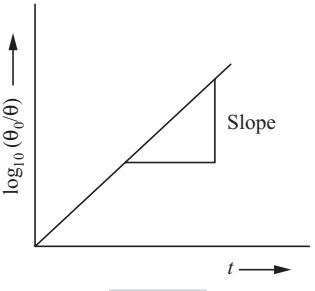


FIG. 12.9

Example 1. *If the charge on a capacitor of capacitance 2 μF in leaking through a high resistance of 100 megaohms is reduced to half its maximum value, calculate the time of leakage.*

Solution. $$R = \frac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$$

Here, $C = 2 \times 10^{-6}$ F, $R = 10^8$ Ω, $Q_0/Q = 2$.

$$\therefore \quad t = 2.3026\, CR \log_{10} (Q_0/Q)$$

$$= 2.3026\, (2 \times 10^{-6})\, 10^8 \log_{10} 2 = 138.7 \text{ s.}$$

Example 2. *In an experiment to determine high resistance by leakage, a capacitor of 0.2 μF is used. It is first fully charged and discharged through a B.G. The observed kick was 12 cm on the scale. The capacitor was fully charged again and allowed to leak through R for 2 sec. The remaining charge in C gave a kick of 6 cm on the same scale when discharged through the B.G. Calculate R.*

Solution. $R = \dfrac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$

$$= \frac{2}{2.3026 \times (0.2 \times 10^{-6}) \log_{10} (12/6)} = 14.43 \text{ megaohms.}$$

12.5 CHARGING AND DISCHARGING OF A CAPACITOR THROUGH AN INDUCTANCE

(*i*) Charging of a capacitor through an inductance

A capacitance *C* and an inductance *L* are joined to a battery through a morse key (Fig. 12.10).

- When the key is pressed, a momentary current flows through the inductance *L* and the capacitor starts gettting charged.
- *E* is the emf of the cell.

Q is the charge on the capacitor at any instant *t*.

The emf equation of the circuit is

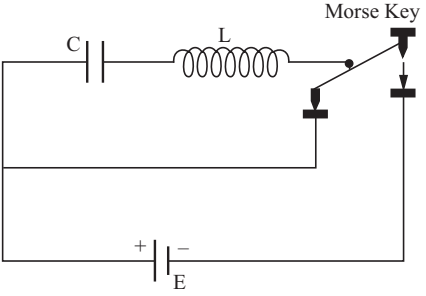


FIG. 12.10

$$L\frac{dI}{dt} + \frac{Q}{C} = E \qquad ...(1)$$

But $$I = \frac{dQ}{dt} \text{ and } \frac{dI}{dt} = \frac{d^2Q}{dt^2}$$

$\therefore$ $$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = E$$

or $$\frac{d^2Q}{dt^2} + \frac{Q - CE}{LC} = 0$$

OR $$\frac{d^2Q}{dt^2} + \frac{1}{LC}(Q - Q_0) = 0 \qquad ...(2)$$

($\because$ $CE = Q_0$, the maximum charge associated with the capacitor).

Eq. (2) is of the form $d^2x/dt^2 + p^2x = 0$ which is a differential equation for undamped simple harmonic motion of period,

$$T = 2\pi/p$$

The time period of undamped oscillations is given by

$$T = \frac{2\pi}{\sqrt{\dfrac{1}{LC}}} = 2\pi\sqrt{LC} \qquad ...(3)$$

Thus, the charging is oscillatory and the frequency of oscillation is given by

$$v = \frac{1}{2\pi\sqrt{LC}} \qquad ...(4)$$

(*ii*) Discharge of a Capacitor through an Inductance

A capacitor C and inductance L are joined to a cell through a morse key (Fig. 12.11). The capacitor is initially charged to its maximum value Q_0 by pressing the morse key (with inductance out of the circuit).

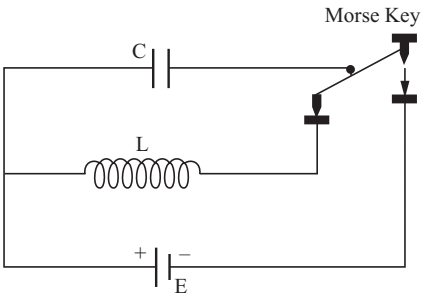


FIG. 12.11

- When the key is now released, the capacitor discharges itself through the inductance L.
- Let I be the current and Q charge left on the capacitor at any instant t.
- There is no external source of emf during discharge.

The equation of emf's is

$$\frac{Q}{C} + L\frac{dI}{dt} = 0$$

or
$$\frac{Q}{C} + L\frac{d^2Q}{dt^2} = 0 \qquad \left[\because \quad I = \frac{dQ}{dt} \text{ and } \frac{dI}{dt} = \frac{d^2Q}{dt^2}\right]$$

$\therefore$
$$\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0 \qquad ...(1)$$

This is the differential equation representing undamped simple harmonic motion. Taking $\frac{1}{LC} = \omega^2$, we have

$$\frac{d^2Q}{dt^2} + \omega^2 Q = 0$$

The general solution of this equation is

$$Q = A \sin(\omega t + \theta), \qquad ...(2)$$

Here, $\omega = \frac{1}{\sqrt{LC}}$

When, $t = 0$, $Q = Q_0$ $\quad\therefore\quad$ $Q_0 = A \sin\theta$.

When $\theta = \pi/2$, $A = Q_0$.

$\therefore$ Eq. (2), becomes

$$Q = Q_0 \sin\left(\frac{t}{\sqrt{LC}} + \frac{\pi}{2}\right) \qquad ...(3)$$

Thus, Q decreases sinusoidally or the discharge is oscillatory.

The natural frequency v of the circuit is given by

$$v = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad ...(4)$$

Importance in wireless telegraphy. The discharge of capacitor through an inductance is oscillatory and simple harmonic in nature.

- When the capacitor is charged, the entire energy of the circuit is stored in the capacitor as electrostatic energy (Fig. 12.12 (*a*)).
- During discharge of the capacitor, a current flows through the inductance coil for a very small time. This sets up a magnetic field within the coil and the whole of the energy is magnetic. (Fig. 12.12 (*b*)).
- When the oscillations are set up, the energy in the circuit is stored alternately in the electric field of capacitor and the magnetic field of the coil.

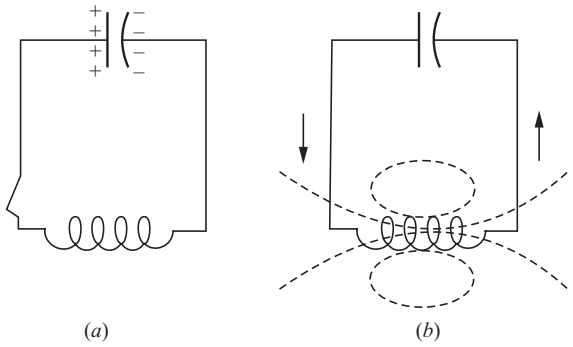


FIG. 12.12

- When the resistance of the circuit is low, a large number of oscillations take place per second. The changes in electric and magnetic fields give rise to electromagnetic radiations, which are propagated through space with the velocity of light.
  The frequency of these oscillations is given by

$$v = \frac{1}{2\pi\sqrt{LC}}.$$

- These electromagnetic oscillations form the basis of wireless telegraphy. With the help of codes, the messages can be transmitted from one place to another.

12.6 GROWTH OF CHARGE IN A CIRCUIT WITH INDUCTANCE, CAPACITANCE AND RESISTANCE

Consider a circuit containing an inductance L, capacitance C and resistance R joined in series to a cell of emf E (Fig. 12.13). When the key K is pressed, the capacitor is charged. Let Q be the charge on the capacitor and I the current in the circuit at an instant t during charging. Then, the p.d. across the capacitor is Q/C and the self induced emf in the inductance coil is $L\,(dI/dt)$, both being opposite to the direction of E. The p.d. across the resistance R is RI.

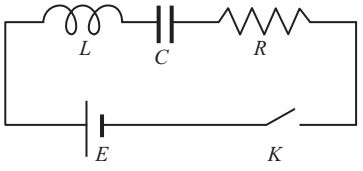


FIG. 12.13

The equation of emf's is

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = E \qquad \ldots(1)$$

But $$I = \frac{dQ}{dt} \text{ and } \frac{dI}{dt} = \frac{d^2Q}{dt^2}$$

$\therefore$ $$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E$$

or $$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q - CE}{LC} = 0$$

Putting $\frac{R}{L} = 2b$ and $\frac{1}{LC} = k^2$, we have

$$\frac{d^2Q}{dt^2} + 2b\frac{dQ}{dt} + k^2\,(Q - CE) = 0 \qquad \ldots(2)$$

Let $x = Q - CE$. Then $\frac{dx}{dt} = \frac{dQ}{dt}$ and $\frac{d^2x}{dt^2} = \frac{d^2Q}{dt^2}$

Eq. (2) becomes, $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + k^2x = 0$ …(3)

Hence the most general solution of Eq. (3) is

$$x = Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$$

Now, $CE = Q_0$= final steady charge on the capacitor.

$\therefore$ $x = Q - CE = Q - Q_0$

Hence $Q - Q_0 = Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$

or $Q = Q_0 + Ae^{[-b+\sqrt{(b^2-k^2)}]t} + Be^{[-b-\sqrt{(b^2-k^2)}]t}$ …(4)

Using initial conditions:

at $t = 0, Q = 0$

$\therefore$ $0 = Q_0 + (A + B)$ or $A + B = -Q_0$ …(5)

$$\frac{dQ}{dt} = A(-b+\sqrt{(b^2-k^2)})\,e^{[-b+\sqrt{(b^2-k^2)}]t} + B[-b-\sqrt{(b^2-k^2)}]\,e^{[-b-\sqrt{(b^2-k^2)}]t}$$

At $t = 0, \frac{dQ}{dt} = 0$

$$0 = A[-b+\sqrt{(b^2-k^2)}] + B[-b-\sqrt{(b^2-k^2)}]$$

$$\sqrt{(b^2-k^2)}\,[A-B] = b(A+B) = -bQ_0$$

or $A - B = -\frac{Q_0 b}{\sqrt{(b^2-k^2)}}$ …(6)

Solving Eqs. (5) and (6),

$$A = -\frac{1}{2}Q_0\left(1+\frac{b}{\sqrt{(b^2-k^2)}}\right) \quad \text{…(7)}$$

$$B = -\frac{1}{2}Q_0\left(1-\frac{b}{\sqrt{(b^2-k^2)}}\right) \quad \text{…(8)}$$

Substituting the values of A and B in Eq. (4), we have

$$Q = Q_0 - \frac{1}{2}Q_0 e^{-bt}\left[\left(1+\frac{b}{\sqrt{(b^2-k^2)}}\right)e^{\sqrt{(b^2-k^2)}\,t} + \left(1-\frac{b}{\sqrt{(b^2-k^2)}}\right)e^{-\sqrt{(b^2-k^2)}\,t}\right] \quad \text{…(9)}$$

Case I. If $b^2 > k^2$, $\sqrt{(b^2-k^2)}$ is real. The charge on the capacitor grows exponentially with time and attains the maximum value Q_0 asymptotically, (curve 1 of Fig. 12.14). The charge is known as *over damped or dead beat*.

Case II. If $b^2 = k^2$, the charge rises to the maximum value Q_0 in a short time (curve 2 of Fig. 12.14). Such a charge is called *critically damped*.

Case III. If $b^2 < k^2$, $\sqrt{(b^2-k^2)}$ is imaginary.

Let $\sqrt{(b^2-k^2)} = i\omega$ where $i = \sqrt{-1}$ and $\omega = \sqrt{k^2-b^2}$

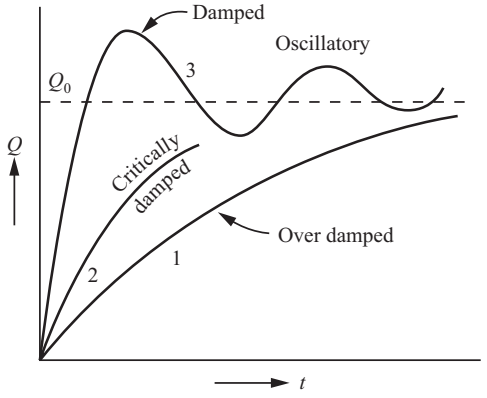


FIG. 12.14

Eq. (9) may be written as

$$Q = Q_0 - \frac{1}{2} Q_0 \, e^{-bt} \left[\left(1 + \frac{b}{i\omega} \right) e^{i\omega t} + \left(1 - \frac{b}{i\omega} \right) e^{-i\omega t} \right]$$

$$Q = Q_0 - Q_0 \, e^{-bt} \left[\frac{e^{i\omega t} + e^{-i\omega t}}{2} + \frac{b}{\omega} \frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right]$$

$$Q = Q_0 - Q_0 \, e^{-bt} \left(\cos \omega t + \frac{b}{\omega} \sin \omega t \right)$$

$$Q = Q_0 \left[1 - \frac{e^{-bt}}{\omega} (\omega \cos \omega t + b \sin \omega t) \right]$$

Let $\omega = k \sin \alpha$ and $b = k \cos \alpha$ so that $\tan \alpha = \omega / b$.

$$Q = Q_0 \left[1 - \frac{e^{-bt}}{\omega} (k \sin \alpha \cos \omega t + k \cos \alpha \sin \omega t) \right]$$

or

$$Q = Q_0 \left[1 - \frac{k e^{-bt}}{\omega} \sin (\omega t + \alpha) \right] \qquad \ldots(10)$$

$$Q = Q_0 \left[1 - \frac{e^{-\frac{R}{2L} t} \sqrt{\frac{1}{LC}}}{\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} \cdot \sin \left[\left(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \right) t + \alpha \right] \right]$$

This equation represents a damped oscillatory charge as shown by the curve (3). The charge oscillates above and below Q_0 till it finally settles down to Q_0 value. The frequency of oscillation in the circuit is given by

$$v = \frac{\omega}{2\pi} = \frac{\sqrt{k^2 - b^2}}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

When $R = 0$, $v = \dfrac{1}{2\pi\sqrt{LC}}$

12.7 DISCHARGE OF A CAPACITOR THROUGH AN INDUCTOR AND A RESISTOR IN SERIES (DECAY OF CHARGE IN LCR CIRCUIT)

Consider a circuit containing a capacitor of capacitance C, an inductance L and a resistance R joined in series (Fig. 12.15). E is a cell. K_2 is kept open. The capacitor is charged to maximum charge Q_0 by closing the key K_1. On opening K_1 and closing key K_2, the capacitor discharges through the inductance L and resistance R. Let I be the current in the circuit and Q be the charge in the capacitor at any instant during discharge. The circuit equation then is

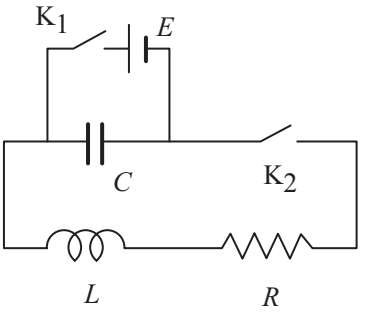


FIG. 12.15

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = 0$$

But, $I = \dfrac{dQ}{dt}$ and $\dfrac{dI}{dt} = \dfrac{d^2Q}{dt^2}$

$\therefore$ $$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0$$

$$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q}{LC} = 0 \quad \ldots(1)$$

Let $\dfrac{R}{L} = 2b$ and $\dfrac{1}{LC} = k^2$, then

$$\frac{d^2Q}{dt^2} + 2b\frac{dQ}{dt} + k^2Q = 0 \quad \ldots(2)$$

The general solution of this equation is

$$Q = A\,e^{(-b+\sqrt{b^2-k^2})t} + Be^{(-b-\sqrt{b^2-k^2})t} \quad \ldots(3)$$

where A and B are arbitrary constants.

When $t = 0$, $Q = Q_0$ and from Eq. (3), $A + B = Q_0$ …(4)

$$\frac{dQ}{dt} = A\,(-b+\sqrt{b^2-k^2})\,e^{(-b+\sqrt{b^2-k^2})t} + B\,(-b-\sqrt{b^2-k^2})\,e^{(-b-\sqrt{b^2-k^2})t}$$

When, $t = 0$, $\dfrac{dQ}{dt} = 0$

$\therefore$ $$A\,(-b+\sqrt{b^2-k^2}) + B\,(-b-\sqrt{b^2-k^2}) = 0$$

$$-b\,(A+B) + \sqrt{b^2-k^2}\,(A-B) = 0$$

$$-bQ_0 + \sqrt{b^2-k^2}\,(A-B) = 0$$

$\therefore$ $$A - B = \frac{bQ_0}{\sqrt{b^2-k^2}} \quad \ldots(5)$$

From Eqs. (4) and (5), we get

$$A = \frac{1}{2}Q_0\left(1 + \frac{b}{\sqrt{(b^2-k^2)}}\right) \text{ and } B = \frac{1}{2}Q_0\left(1 - \frac{b}{\sqrt{(b^2-k^2)}}\right)$$

Putting these values of A and B in Eq. (3), we get

$$Q = \frac{1}{2} Q_0 e^{-bt} \left[\left(1 + \frac{b}{\sqrt{(b^2 - k^2)}} \right) e^{\sqrt{(b^2 - k^2)}\, t} + \left(1 - \frac{b}{\sqrt{(b^2 - k^2)}} \right) e^{-\sqrt{(b^2 - k^2)}\, t} \right] \quad \ldots(6)$$

Case I. If $b^2 > k^2, \sqrt{(b^2 - k^2)}$ is real and positive and the charge of the capacitor decays exponentially, becoming zero asymptotically (curve 1 of Fig. 12.16). This discharge is known as *over damped, non-oscillatory or dead beat*.

Case II. When $b^2 = k^2$, $Q = Q_0 (1 + bt)\, e^{-bt}$

This represents a *non-oscillatory* discharge. This discharge is known as *critically damped* (Curve 2 of Fig. 12.16). The charge decreases to zero exponentially in a short time.

Case III. If $b^2 < k^2, \sqrt{(b^2 - k^2)}$ is imaginary.

$$\sqrt{(b^2 - k^2)} = i\omega, \qquad \text{where } \omega = \sqrt{(k^2 - b^2)}$$

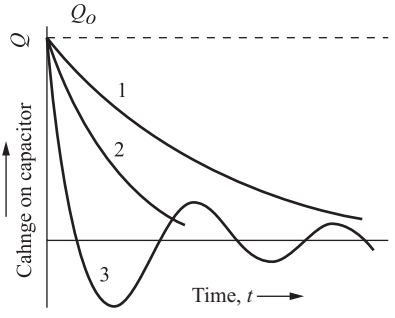


FIG. 12.16

$$\therefore \quad Q = \frac{1}{2} Q_0\, e^{-bt} \left[\left(1 + \frac{b}{i\omega} \right) e^{i\omega t} + \left(1 - \frac{b}{i\omega} \right) e^{-i\omega t} \right]$$

$$= Q_0 e^{-bt} \left[\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) + \frac{b}{\omega} \left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) \right]$$

$$= \frac{Q_0 e^{-bt}}{\omega} (\omega \cos \omega t + b \sin \omega t)$$

Let $\omega = k \sin \alpha$ and $b = k \cos \alpha$ so that $\tan \alpha = \dfrac{\omega}{b}$.

$$Q = \frac{Q_0 e^{-bt} k}{\omega} (\cos \omega t \sin \alpha + \cos \alpha \sin \omega t)$$

$$= \frac{Q_0 e^{-bt} k}{\omega} \sin (\omega t + \alpha)$$

$$Q = \frac{Q_0 e^{-\frac{R}{2L} t}}{\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2} \right) \sqrt{LC}}} \sin \left(\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2} \right)} t + \alpha \right) \quad \ldots(7)$$

This equation represents a *damped oscillatory* charge as shown by the curve (3). The charge oscillates above and below zero till it finally settles down to zero value.

The frequency of oscillation in the circuit is given by

$$v = \frac{\omega}{2\pi} = \frac{\sqrt{k^2 - b^2}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

When $R = 0$, $v = \dfrac{1}{2\pi\sqrt{LC}}$

The condition for oscillatory discharge is

$$\frac{R^2}{4L^2} < \frac{1}{LC} \quad \text{or} \quad R < 2\sqrt{\frac{L}{C}}$$

Importance in Wireless Telegraphy. The discharge of a capacitor through an inductance is oscillatory if the resistance R of the circuit is less than $2\sqrt{(L/C)}$. During the discharge, the energy of the charged capacitor is stored in the magnetic field produced in the inductance coil, then again back in the electric field between the capacitor plates, and so on. If the fields are caused to alternate rapidly, some energy escapes from the circuit permanently in the form of electro-magnetic waves which travel through space with the speed of light. These waves form the basis of wireless telegraphy.

Messages can be transmitted from one place to another with the help of codes.

Example 1. *If a battery, of emf 100 volts, is connected in series with an inductance of 10 mH, a capacitor of 0.05* μ*F and a resistance of 100* Ω*, find (i) the frequency of the oscillatory current, (ii) the logarithmic decrement and (iii) the final capacitor charge.*

Solution. $\dfrac{R^2}{4L^2} = \dfrac{100^2}{4 \times (10^{-2})^2} = 2.5 \times 10^7$

and $\dfrac{1}{LC} = \dfrac{1}{10^{-2} \times 0.05 \times 10^{-6}} = 2 \times 10^9$

$R^2/4L^2 < 1/LC$. *Hence the charging of the capacitor is oscillatory.*

$$v = \frac{1}{2\pi}\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)} = \frac{1}{2 \times 3.14}\sqrt{(2 \times 10^9 - 2.5 \times 10^7)} = 7076.6 \text{ Hz.}$$

The logarithmic decrement $= \dfrac{RT}{2L} = \dfrac{100}{2 \times 10^{-2} \times 7076.6} = 0.707.$

The final capacitor charge $Q_0 = EC = 100 \times 0.05 \times 10^{-6} = 5\mu C$.

Example 2. *A charged capacitor of capacitance 0.01* μ*F is made to discharge through a circuit consisting of a coil of inductance 0.1 henry and an unknown resistance. What should be the maximum value of the unknown resistance, if the discharge of the capacitor is to be oscillatory?*

Sol. If R is the maximum value of the resistance for the discharge to be oscillatory, then

$$\frac{R^2}{4L^2} = \frac{1}{LC} \text{ or } R = \sqrt{\frac{4L}{C}} = \sqrt{\left(\frac{4 \times 0.1}{0.01 \times 10^{-6}}\right)} = 6324\,\Omega$$

Example 3. *(i) Find out whether the discharge of a capacitor through a circuit containing the following elements, is oscillatory.*

$$C = 0.2\ \mu F, \quad L = 10\ mH, \quad R = 250\ ohm$$

(*ii*) *If so, find the frequency.*

(*iii*) *Calculate the maximum value of the resistance possible so as to make the discharge oscillatory.*

Solution. (*i*) The condition for oscillations is that

$$\frac{R^2}{4L^2} < \frac{1}{LC}$$

or $$R^2 < \frac{4L}{C}, \quad \text{or} \quad R < \sqrt{\frac{4L}{C}}$$

We have $$\sqrt{\frac{4L}{C}} = \sqrt{\frac{4\times10\times10^{-3}}{0.2\times10^{-6}}} = 447 \text{ ohm}$$

It is given that $R = 250$ ohm

$\therefore$ $$R < \sqrt{\frac{4L}{C}}$$

Therefore the discharge is oscillatory.

(*ii*) The frequency of oscillations is given by

$$v = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{1}{10\times10^{-3}\times0.2\times10^{-6}} - \frac{(250)^2}{4\times(10\times10^{-3})^2}}$$

$$= 54\times10^6 \text{ Hz.}$$

(*iii*) The maximum possible resistance of oscillatory discharge is given by the equation

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

$$R = \sqrt{\frac{4L}{C}} = 447 \text{ ohm.}$$

EXERCISE

1. The inductive time constant is:

(*a*) $\frac{L}{R}$ (*b*) T (*c*) CR (*d*) LR

(Kumaun 2015)

2. If $C = 0.1$ μF and $R = 100$ mega ohm, the time constant for the circuit will be:

(*a*) 1 sec (*b*) 10 sec (*c*) 100 sec (*d*) 1000 sec

(Kumaun 2011, 2014)

3. Derive an expression for growth of current through inductance L and resistance R in a circuit. **(Kumaun 2014)**

4. Analyse the growth and decay of current in an *LR* circuit. **(Kumaun 2011)**

5. Define time constant of *L–R* circuit. What are its dimensions? **(Kanpur 2003, Rohilkhand 2004)**

6. Discuss the charging and discharging of a capacitor through resistance. What is time constant? **(Gorakhpur 85, Rohilkhand 83)**

7. Discuss with relevant theory the method of determining a high resistance by the method of leakage. **(Indore 83, Lucknow 84)**
8. Prove that when a capacitor is discharged through an inductance of negligible resistance, the discharge is oscillatory and simple harmonic. Calculate its periodic time and frequency.
9. A charged condenser discharges through an inductance and a resistance. Discuss the nature of the discharge with special reference to oscillatory discharge. **(Kumaun 2012)**
10. A condenser of capacity 0.1 μF connected to a resistance of 10 mega ohms is charged to a certain potential and then isolated. Find the time the potential will take to fall to half its original value (given ln2 = 0.693) **(Kumaun 2011)**
11. Show that capacitive time constant τ_C has the dimensions of time.
12. A resistance R and an inductance L are connected to a battery of E volts. When will the potential difference across inductance be equal to that across resistance? **(Kanpur 84)**
[**Hints:** Instantaneous pd across resistance = $RI = RI_0\,[1 - e^{-(R/L)t}]$

Instantaneous pd across inductance = $L\dfrac{dI}{dt} = RI_0\, e^{-(R/L)t}$

$\therefore \quad 1 - e^{-(R/L)t} = e^{-(R/L)t}$

or $\quad t = \dfrac{L}{R}\log_e 2 = 0.69\dfrac{L}{R}$]
13. A 0.5 μ**F** capacitor is discharged through a resistance of 10 megohm. Find the time taken for half the charge on the capacitor to escape. (Given $\log_e 2 = 0.69$) **(Kumaun 2012)**

ANSWERS

1. (*a*) **2.** (*b*)

10. 0.693s

11. $\tau_C = CR$. Now, capacitance $C = \dfrac{Q}{V} = \dfrac{It}{V} = \dfrac{t}{(V/I)} = \dfrac{t}{R}$

$$\tau_C = \frac{t}{R} \times R = t$$

$\therefore$ τ_C has dimensions of time.

13. $Q = Q_0^{-t/CR}$ or $\dfrac{Q}{Q_0} = e^{-t/CR}$

$\therefore \quad \dfrac{1}{2} = e^{-t/(0.5 \times 10^{-6} \times 10 \times 10^{6})}$

or $\quad e^{t/5} = 2$

$$\frac{t}{5} = \log_e 2 = 0.69$$

$\therefore \quad t = 5 \times 0.69 = \mathbf{3.45\ sec}$.

13

CHAPTER

Alternating Current

Learning Objectives

After reading this chapter, you should be able to:

- Distinguish between the mean value, peak value and the root mean square value of an alternating current and establish the relation between them.
- Explain the method of solving alternating current problems with the aid of complex quantities. Investigate by such a method the behaviour of a resonant circuit consisting of a capacitance *C* in parallel with a coil of inductance *L* and resistance *R*.
- Discuss series and parallel resonance circuits. Explain their uses.
- Explain in detail the construction, theory and working of a transformer.
- Explain how a rotating magnetic field is produced and show how it can be used in the construction of a motor.
- Discuss an accurate A.C. bridge method for the determination of self-inductance of a coil.

13.1 EMF INDUCED IN A COIL ROTATING IN A MAGNETIC FIELD

Consider a rectangular coil of N turns and of length a and width b rotating with uniform angular velocity ω about its axis in a uniform magnetic field **B** (Fig. 13.1).

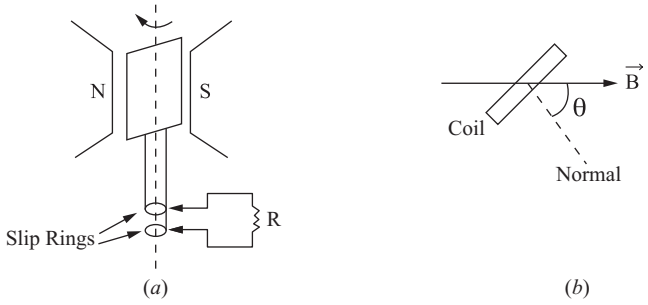


FIG. 13.1

The axis of rotation is at right angles to the field. As the coil rotates, the magnetic flux passing through it changes. Hence an emf is induced in the coil.

Suppose we start timing from the instant when the plane of the coil is at right angles to the field **B**, *i.e.*, when the angle between the normal to the plane of the coil and the direction of the field is zero. Then, at an instant t, the normal to the plane of the coil will make angle θ (= ωt) with the direction of **B**.

The magnetic flux linked with N turns of the coil is

$$\phi = NBA \cos\theta = NBA \cos\omega t.$$

where A (= ab) is the area of the coil.

The instantaneous induced emf,

$$E = -\frac{d\phi}{dt} = -\frac{d}{dt}(NBA \cos\omega t) = NBA\,\omega \sin\omega t = E_0 \sin\omega t$$

Here, $E_0 = NBA\omega$, called the *peak value* of the e.m.f.

Now, $\omega = 2\pi v$ where v = frequency of alternating voltage.

Thus, when $\omega t = 0$, $\sin\omega t = 0$ and $E = 0$

$\omega t = \pi/2$, $\sin\omega t = 1$ and $E = E_0$

$\omega t = \pi$, $\sin\omega t = 0$ and $E = 0$

$\omega t = 3\pi/2$, $\sin\omega t = -1$, and $E = -E_0$

and $\omega t = 2\pi$, $\sin\omega t = 0$ and $E = 0$ again,

A graph of E against ωt is a sine curve (Fig. 13.2). Such an e.m.f. is called an 'alternating e.m.f.' The resulting current in the coil, if the coil is part of a *closed circuit*, is the 'alternating current.'

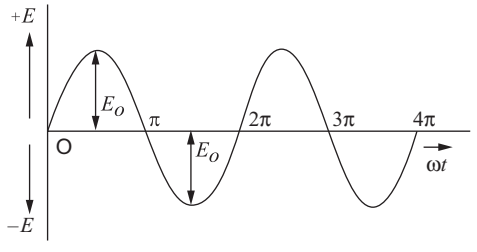


FIG. 13.2

The corresponding current I through the circuit is given by

$$I = I_0 \sin \omega t.$$

The time of one cycle is known as *time period T*, the number of cycles per second the frequency ν (= $1/T$), the peak value of current or voltage the *amplitude*.

Fig. 13.3 shows one cycle of ac voltage of current.

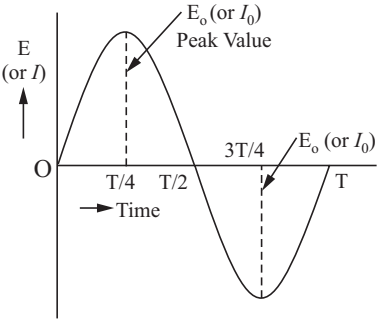


FIG. 13.3

Peak Value of Alternating Current or emf. *The maximum value of alternating current or emf in the positive or negative direction is called peak value of alternating current or emf. It is denoted by* I_0 or E_0.

Mean Value of Alternating Current. *Mean value of alternating current is defined as its average over half a cycle.*

$$I_{mean} = \frac{\int_0^{T/2} 1\, dt}{T/2} = \frac{\int_0^{\pi/\omega} I_0 \sin \omega t\, dt}{\pi/\omega}$$

$$= \frac{I_0 \omega}{\pi}\left[\frac{-\cos \omega t}{\omega}\right]_0^{\pi/\omega}$$

$$= -\frac{I_0}{\pi}[\cos \pi - \cos 0]$$

$$= \frac{2I_0}{\pi} = 0.637 I_0$$

Similarly, $E_{mean} = 0.637\, E_0$

Root mean square value of an alternating current. *It is defined as the square root of the average of* I^2 *during a complete cycle.*

$$\bar{I^2} = \frac{\int_0^{2\pi/\omega} I^2\, dt}{2\pi/\omega} = \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2 \omega t\, dt}{2\pi/\omega}$$

$$= \frac{I_0^2\, \omega}{2\pi}\int_0^{2\pi/\omega} \frac{1}{2}(1 - \cos 2\omega t)\, dt$$

$$= \frac{I_0^2 \omega}{4\pi}\left[t - \frac{\sin 2\omega t}{2\omega}\right]_0^{2\pi/\omega}$$

$$= \frac{I_0^2 \omega}{4\pi}\left[\frac{2\pi}{\omega}\right] = \frac{I_0^2}{2}$$

$$I_{rms} = \sqrt{\bar{I^2}} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

Similarly, $E_{rms} = \frac{E_0}{\sqrt{2}} = 0.707\, E_0$

Form factor. The form factor gives an indication of the wave shape of the alternating voltage or current. *It is defined as the ratio of the virtual or r.m.s. value to the average value of alternating current or voltage.* Thus in the case of a sinusoidal current (or voltage), form factor is,

$$\text{Form factor} = \frac{I_{r.m.s}}{I_{mean}} = \frac{E_{r.m.s}}{E_{mean}} = \frac{0.707\, E_0}{0.637\, E_0} = 1.11$$

Effective value or virtual value of an A.C. *The rms value of an alternating current can also be defined as that direct current which produces the same rate of heating in a given resistance. Therefore, the r.m.s. value of alternating current is also called as the 'effective' or the 'virtual' value of the current.*

$$I_{virtual} = \frac{I_0}{\sqrt{2}} = I_{rms}$$

Suppose an alternating current of instantaneous value $I = I_0 \sin \omega t$ is flowing through a circuit of resistance R.

$$\left.\begin{matrix}\text{Total quantity of heat produced} \\ \text{over the complete cycle}\end{matrix}\right\} = H = \int_0^T I^2 \, R \, dt \qquad \ldots(1)$$

Let I_v stand for the root mean square or virtual value of the current. Then the heat produced in time T is given by

$$H = I_v^2 RT \qquad \ldots(2)$$

Comparing Eqs. (1) and (2),

$$I_v^2 \, RT = \int_0^T I^2 R \, dt \quad \text{or} \quad I_v^2 T = \int_0^T I^2 dt$$

or
$$I_v^2 T = \int_0^T I_0^2 \sin^2 \omega t \, dt = \frac{I_0^2}{2} \int_0^T 2 \sin^2 \omega t \, dt$$

$$= \frac{I_0^2}{2} \int_0^T (1 - \cos 2\omega t) \, dt = \frac{I_0^2}{2} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

$$= \frac{I_0^2 \cdot T}{2}$$

or
$$I_v^2 = \frac{I_0^2}{2}$$

or
$$I_v = \frac{I_0}{\sqrt{2}} = 0.707 \, I_0$$

Similarly, the r.m.s. value of an alternating voltage can be defined as that direct voltage which produces the same rate of heating, in a given resistance. The r.m.s. value of alternating voltage is also called as the 'effective' or the 'virtual' value of the voltage.

$$E_{virtual} = \frac{E_0}{\sqrt{2}} = E_{rms}$$

1. Mean or average value of A.C voltage for half cycle.

The alternating e.m.f is represented as,

$$E = E_0 \sin \omega t.$$

Mean value of alternating e.m.f over one half cycle is,

$$E_{av} = \frac{\int_0^{T/2} E_0 \sin \omega t \, dt}{\int_0^{T/2} dt} = \frac{E_0}{T/2} \left[\frac{-\cos \omega t}{\omega} \right]_0^{T/2}$$

$$= \frac{2E_0}{T} \times \frac{T}{2\pi} \left[-\cos \left(\frac{2\pi}{T} \right) t \right]_0^{T/2} \qquad \left(\because \quad \omega = \frac{2\pi}{T} \right)$$

$$= \frac{E_0}{\pi}[-\cos\pi + \cos 0]$$

$$\therefore \quad E_{av} = \frac{2E_0}{\pi}$$

2. R.M.S. value of Alternating Voltage

The R.M.S value of an alternating e.m.f is defined as that constant direct e.m.f which when applied to a resistor will develop the same heat energy in it as the given alternating e.m.f during the same interval of time.

The root mean square (r.m.s.) value of an alternating e.m.f is the square root of the mean of the squares of the emf taken during one cycle.

Instantaneous value of the e.m.f. = $E = E_0 \sin \omega t$

$$\text{Mean square value} = \frac{\int_0^T E_0^2 \sin^2 \omega t \, dt}{\int_0^T dt} = \frac{1}{T}\int_0^T E_0{}^2 \sin^2 \omega t \, dt$$

$$\int_0^T \sin^2 \omega t \, dt = \int_0^T \frac{1}{2}(1 - \cos 2\omega t)\, dt = \int_0^T \frac{1}{2}\, dt - \frac{1}{2}\int_0^T \cos 2\omega t \, dt$$

$$= \frac{1}{2}[t]_0^T - \frac{1}{2}\left[\frac{\sin 2\omega t}{2\omega}\right]_0^T = \frac{T}{2}$$

$$\text{Mean square value} = \frac{1}{T} \times E_0^2 \times \frac{T}{2} = \frac{E_0^2}{2}$$

$$\therefore \quad E_{rms} = E_0/\sqrt{2}$$

Impedance, Reactance and Admittance.

- Alternating current in a circuit is controlled by resistance, inductance and capacitance.

Impedance (Z). In alternating current circuit, the ratio of e.m.f. applied and consequent current produced is called the *impedance* and is denoted by Z

i.e., $$Z = \frac{E}{I}.$$

In any circuit, the ratio of the effective voltage to the effective current is defined as the impedance Z of the circuit.

Physically impedance of ac circuit is the hindrance offered by the circuit to the flow of ac through it.

Reactance (X): The hindrance offered by inductance and capacitance to the flow of ac in an ac circuit is called *reactance* and is denoted by X. Thus when there is no ohmic resistance in the circuit, the reactance is equal to impedance. The reactance due to inductance alone is called inductive reactance and is denoted by X_L. The reactance due to capacitance alone is called the capacitive reactance and is denoted by X_C.

Admittance. The inverse of impedance is called the admittance and is denoted by Y *i.e.*,

$$Y = \frac{1}{Z}$$

13.2 ALTERNATING CURRENT (A.C.) CIRCUIT THEORY

(*i*) A.C. Circuit Containing Resistance only

Suppose an alternating sinusoidal voltage

$$E = E_0 e^{j\omega t}$$

is applied across a pure resistance R (Fig. 13.4).

The current through the resistance at any instant t is,

$$I = \frac{E}{R} = \frac{E_0 e^{j\omega t}}{R} = I_0 e^{j\omega t}$$

Here, $$I_0 = \frac{E_0}{R}.$$

Actual value of current = Imaginary part of $I_0\, e^{j\omega t} = I_0 \sin \omega t$.

$\therefore$ $$I = I_0 \sin \omega t$$

Hence the current is in phase with the driving voltage (Fig. 13.5).

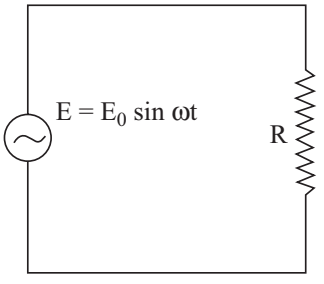


FIG. 13.4

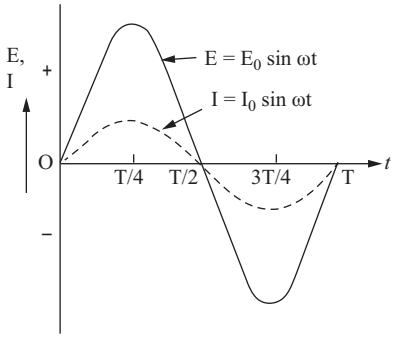


FIG. 13.5

(*ii*) A.C. Circuit Containing Inductance only

Let an alternating voltage

$$E = E_0\, e^{j\omega t}$$

be applied across an inductance L (Fig. 13.6). Let I be the instantaneous current and dI/dt be the rate of change of current with time through the inductance.

The voltage induced across the inductance $= L\dfrac{dI}{dt}$.

$$L\frac{dI}{dt} = E_0\, e^{j\omega t}$$

or $$dI = \frac{E_0}{L} e^{j\omega t}\, dt$$

Integrating, $$I = \frac{E_0}{j\omega L} e^{j\omega t} = \frac{E}{j\omega L} \quad \ldots(1)$$

$\therefore$ $$I = \frac{E_0}{\omega L} e^{j(\omega t - \pi/2)}$$

Let $I_0 = \dfrac{E_0}{\omega L}$ be the maximum current. Then,

$$I = I_0 e^{j(\omega t - \pi/2)}$$

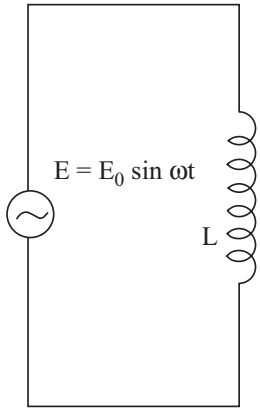


FIG. 13.6

The actual current in the circuit is

$$I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right) \qquad ...(2)$$

In a *pure inductor*, *the current lags behind the emf by* $\pi/2$ (Fig. 13.7).

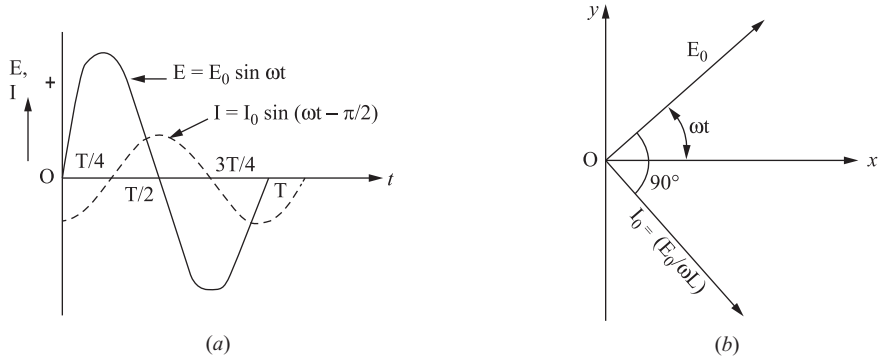


FIG. 13.7

(*iii*) A.C. Circuit Containing Capacitance only

Let an alternating voltage,

$$E = E_0 e^{j\omega t}$$

be applied across the plates of a capacitor of capacitance C (Fig. 13.8). Let q be the charge at any instant across the plates of the capacitor. The p.d. across the plates at any instant must be equal to the applied emf at that instant. Thus

$$\frac{q}{C} = E_0 e^{j\omega t}$$

The current through the circuit is given by,

$$I = \frac{dq}{dt} = \frac{d}{dt}(CE_0\, e^{j\omega t}) = j\omega C\, E_0 e^{j\omega t}$$

or

$$I = \frac{E_0}{1/\omega C} e^{j(\omega t + \pi/2)} = I_0 e^{j(\omega t + \pi/2)}$$

Here, $I_0 = \dfrac{E_0}{1/\omega C}$ is the peak value of alternating current.

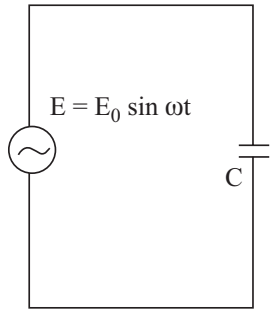


FIG. 13.8

The actual current in the circuit is

$$I = I_0 \sin\left(\omega t + \frac{\pi}{2}\right).$$

In a pure capacitor, *the current leads the emf by* $\pi/2$ (Fig. 13.9)

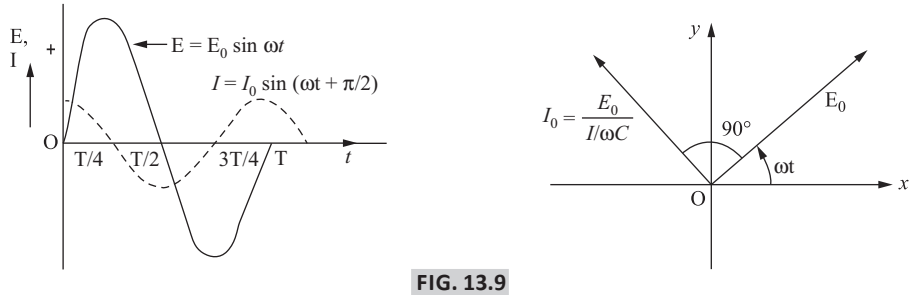


FIG. 13.9

(*iv*) A.C. Circuit Containing Inductance and Resistance in Series

Let an alternating emf

$$E = E_0 e^{j\omega t}$$

be applied to a circuit having an inductance L and a non-inductive resistance R in series (Fig. 13.10).

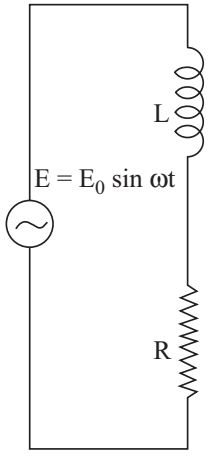


FIG. 13.10

The potential drop across the inductance is

$$V_L = j\omega LI$$

The potential drop across resistance is

$$V_R = RI$$

Here, I is the current at any instant t.

$\therefore$ $$E = j\omega LI + RI$$

Current in the circuit,

$$I = \frac{E}{R + j\omega L} \qquad ...(1)$$

But $$I = \frac{E}{Z} \qquad ...(2)$$

Impedance of R–L circuit,

$$Z = R + j\omega L \qquad ...(3)$$

$\therefore$ $$I = \frac{E_0 e^{j\omega t}}{\sqrt{(R^2 + \omega^2 L^2)} e^{j\theta}} \qquad \left(\text{where } \tan\theta = \frac{\omega L}{R}\right)$$

$$= \frac{E_0}{\sqrt{(R^2 + \omega^2 L^2)}} e^{j(\omega t - \theta)} \qquad ...(4)$$

$$= I_0 e^{j(\omega t - \theta)} \qquad ...(5)$$

Here, $$I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} \qquad ...(6)$$

It represents the peak value of the current through the circuit.

The impedance Z of the circuit is given by the term

$$\sqrt{R^2 + \omega^2 L^2}.$$

The current lags in phase behind the emf by an angle

$$\theta = \tan^{-1}\frac{\omega L}{R}.$$

The variation of instantaneous values of emf and current with time are represented graphically in Fig. 13.11.

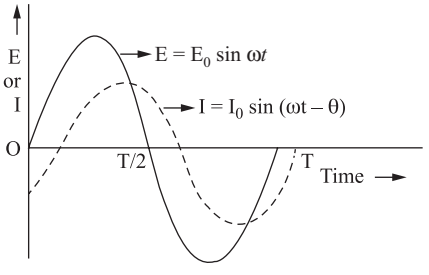


FIG. 13.11

(*v*) A.C. Circuit Containing Capacitance and Resistance in Series

Let an alternating voltage $E = E_0 e^{j\omega t}$ be applied to a circuit containing a resistance R and a capacitor C in series (Fig. 13.12). Let I be the current through the circuit and q, the charge on the plates of the capacitor at any instant. Then,

$$I\left(R + \frac{1}{j\omega C}\right) = E = E_0 e^{j\omega t} \qquad ...(1)$$

$$I = \frac{E}{R + \frac{1}{j\omega C}} \qquad ...(2)$$

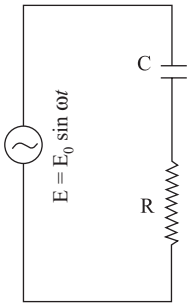


FIG. 13.12

But $$I = \frac{E}{Z} \qquad ...(3)$$

∴ Impedance of series *R*–*C* circuit,

$$Z = R + \frac{1}{j\omega C} = R - \frac{j}{\omega C}$$

$$I = \frac{E}{R - \frac{j}{\omega C}} = \frac{E_0 e^{j\omega t}}{\sqrt{\left[\left(R^2 + \frac{1}{\omega^2 C^2}\right)\right]} e^{-j\theta}} \qquad ...(4)$$

Here, $$\theta = \tan^{-1}\left(\frac{1/\omega C}{R}\right) \qquad ...(5)$$

∴ $$I = \frac{E_0}{\sqrt{\{R^2 + (1/\omega C)^2\}}} e^{j(\omega t + \theta)} \qquad ...(6)$$

The current in this circuit thus leads the applied voltage by an angle θ (Fig. 13.13).

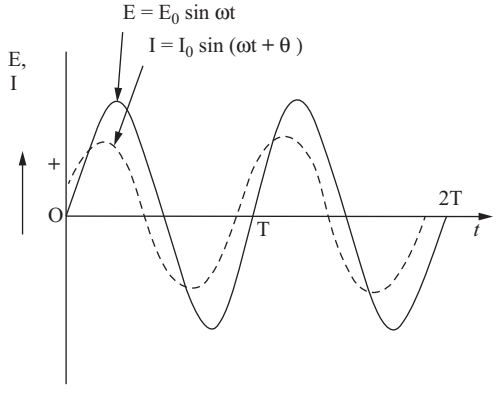


FIG. 13.13

The impedance of series *R*–*C* circuit, $Z = \sqrt{[R^2 + (1/\omega C)^2]}$.

The actual current in the circuit is $I = I_0 \sin(\omega t + \theta)$

(*vi*) Parallel Resonant Circuit

A parallel resonant circuit consisting of an inductance *L* and a capacitance *C* connected in parallel to the alternating e.m.f. is shown in Fig. 13.14.

Let the applied emf be

$$E = E_0 \sin \omega t$$

Current through the inductance lags behind the applied emf by $\frac{\pi}{2}$ and is given by,

$$I_L = \frac{E_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

Here ωL is the reactance of the inductance.

The current through the capacitor leads the applied emf by $\pi/2$ and is given by

$$I_C = \frac{E_o}{(1/\omega C)}\sin\left(\omega t + \frac{\pi}{2}\right)$$

Here, $(1/\omega C)$ is the reactance of the capacitance.

The total current through the circuit is given by

$$I = I_L + I_C$$

$$= \frac{E_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right) + \frac{E_0}{1/\omega C}\sin\left(\omega t + \frac{\pi}{2}\right)$$

$$= -\frac{E_0}{\omega L}\cos\omega t + \frac{E_0}{(1/\omega C)}\cos\omega t$$

or $$I = E_0\left(\omega C - \frac{1}{\omega L}\right)\cos\omega t \qquad ...(1)$$

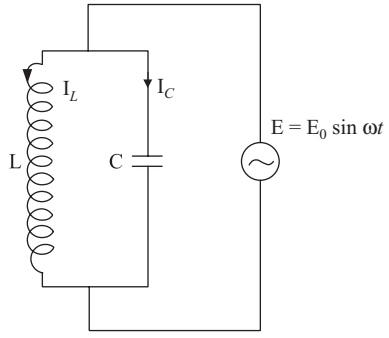


FIG. 13.14

If for a particular frequency v_0, $\omega C = \frac{1}{\omega L}$, then from Eq. (1), the current becomes zero.

The circuit now offers an infinite impedance. Such a circuit which offers an infinite impedance to the A.C. is called a parallel resonant circuit. The frequency is called the resonant frequency.

Thus at v_0, we have,

$$\omega C = \frac{1}{\omega L}$$

or $$\omega = \frac{1}{\sqrt{LC}}$$

or $$2\pi v_0 = \frac{1}{\sqrt{LC}}$$

$\therefore$ $$v_0 = \frac{1}{2\pi\sqrt{LC}} \qquad ...(2)$$

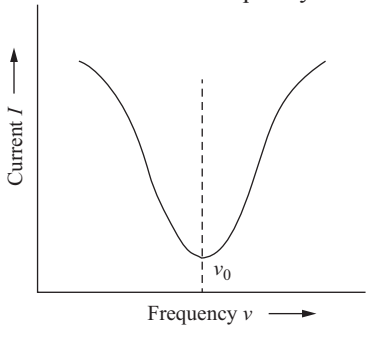


FIG. 13.15

The variation of the current with frequency is shown in Fig. 13.15.

13.3 AC CIRCUIT CONTAINING RESISTANCE, INDUCTANCE AND CAPACITANCE IN SERIES (SERIES RESONANCE CIRCUIT)

Let an alternating emf $E = E_0 \sin \omega t$ be applied to a circuit containing a resistance R, inductance L and capacitance C in series (Fig. 13.16). Let at any instant, I be the current in the circuit and Q be the charge on the capacitor.

The potential drop across the resistance = RI

The E.M.F. induced in the inductance = $L\frac{dI}{dt}$

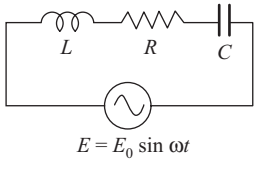


FIG. 13.16

The potential across the plates of the capacitor = Q/C.

$$\therefore \quad L\frac{dI}{dt} + RI + \frac{Q}{C} = E_0 \sin \omega t.$$

Differentiating with respect to t,

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}\frac{dQ}{dt} = E_0\,\omega \cos \omega t \qquad \ldots(1)$$

Let the trial solution be of the form

$$I = I_0 \sin (\omega t - \phi) \qquad \ldots(2)$$

where I_0 and ϕ are constants to be determined.

$$\frac{dI}{dt} = I_0 \omega \cos (\omega t - \phi)$$

and

$$\frac{d^2I}{dt^2} = -I_0\omega^2 \sin (\omega t - \phi)$$

Substituting these value of I, $\frac{dI}{dt}$ and $\frac{d^2I}{dt^2}$, in Eq. (1), we get

$$-LI_0\,\omega^2 \sin (\omega t - \phi) + RI_0\omega \cos (\omega t - \phi) + \frac{I_0}{C}\sin (\omega t - \phi) = E_0\,\omega \cos \omega t$$

or

$$\left(-L\omega^2 + \frac{1}{C}\right) I_0 \sin (\omega t - \phi) + R\omega I_0 \cos (\omega t - \phi)$$

$$= E_0\omega\,[\cos (\omega t - \phi) + \phi]$$

$$= E_0\omega\,[\cos (\omega t - \phi)\cos\phi - \sin(\omega t - \phi)\sin\phi]$$

Equating the coefficients of $\sin (\omega t - \phi)$ and $\cos (\omega t - \phi)$ on either side,

$$\left(-L\omega^2 + \frac{1}{C}\right) I_0 = -E_0\omega \sin \phi, \qquad \ldots(3)$$

and

$$R\omega I_0 = E_0\omega \cos \phi. \qquad \ldots(4)$$

Dividing Eq. (3) by Eq. (4), we get

$$\tan \phi = -\frac{\left(-L\omega^2 + \frac{1}{C}\right)}{R\omega} = \frac{\omega L - \frac{1}{C\omega}}{R} \qquad \ldots(5)$$

Squaring and adding Eqs. (3) and (4), we get

$$I_0^2\left[\left(-L\omega^2 + \frac{1}{C}\right)^2 + R^2\omega^2\right] = E_0^2\omega^2$$

or

$$I_0^2\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right] = E_0^2$$

or

$$I_0 = \frac{E_0}{\sqrt{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]}} \qquad \ldots(6)$$

Substituting the value of I_0 in Eq. (2), we get

$$I = \frac{E_0}{\sqrt{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]}} \sin(\omega t - \phi) \quad \ldots(7)$$

where $\phi = \tan^{-1} \dfrac{\omega L - \dfrac{1}{\omega C}}{R}$

Eq. (7) represents the current at any instant.

The quantity $\sqrt{\left[R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2\right]}$ is the *impedance Z* of the circuit.

$L\omega$ and $1/\omega C$ respectively represent inductive reactance X_L and capacitive reactance X_C. Thus $Z = \sqrt{[R^2 + (X_L - X_C)^2]}$.

The current lags in phase behind e.m.f. by an angle

$$\phi = \tan^{-1} \frac{\omega L - \dfrac{1}{\omega C}}{R} = \tan^{-1} \frac{X_L - X_C}{R}$$

The following three cases arise:

(*i*) When $X_L > X_C$, ϕ is positive so that the *current lags behind the applied emf.*

(*ii*) When $X_L < X_C$, ϕ is negative, so that the *current leads the applied emf.*

(*iii*) When $X_L = X_c$, $\phi = 0$, and the *current is in phase with the emf.*

Series Resonant Circuit

The value of current at any instant in a series *LCR* circuit is given by

$$I = \frac{E_0}{\sqrt{\left\{R^2 + \left(\omega L - \frac{1}{C\omega}\right)^2\right\}}} \sin(\omega t - \phi) \quad \ldots(1)$$

where $\sqrt{\left\{R^2 + \left(\omega L - \dfrac{1}{C\omega}\right)^2\right\}} = Z$

is called the impedance of the circuit.

At a particular frequency, $\omega L = \dfrac{1}{\omega C}$ so that the impedance becomes minimum being given by $Z = R$. *This particular frequency v_0 at which the impedance of the circuit becomes minimum and, therefore the current becomes maximum, is called the* ***resonant frequency*** *of the circuit. Such a circuit which admits maximum current is called series resonant circuit.*

Thus at v_0, we have

$$\omega L = \frac{1}{\omega C} \quad \text{or} \quad 2\pi v_0 L = \frac{1}{2\pi v_0 C}$$

or

$$v_0 = \frac{1}{2\pi\sqrt{LC}}$$

The maximum current in the circuit = $I_0 = E_0/R$. The variation of current with frequency of applied voltage is shown in Fig. 13.17. The sharpness of peak depends upon the resistance R of the circuit. For low resistance, the peak is sharp.

Acceptor Circuit. The series resonant circuit is often called an 'acceptor' circuit. By offering *minimum impedance* to currents at the resonant frequency, it is able to select or accept most readily the current of this one frequency from among those of many frequencies.

In radio receivers, the resonant frequency of the circuit is tuned (by changing C) to the frequency of the signal desired to be detected.

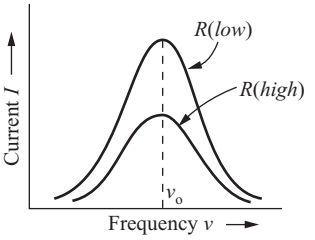


FIG. 13.17

Voltage Magnification. At resonance, the (peak) current through the circuit is

$$I_0 = \frac{E_0}{R}$$

The (peak) voltage across the inductance is

$$V_L = X_L\, I_0 = \omega L \times \frac{E_0}{R} = \frac{\omega L}{R}(E_0) = Q\, E_0,$$

where Q is known as the *quality factor* of the circuit. Thus at resonance the voltage drop across L (which is also the voltage across C) is Q times the applied voltage. Hence the chief characteristic of the series resonant circuit is 'voltage magnification'. This voltage magnification does not increase the power in the circuit as the reactive component of the power is *wattless*.

The Q-factor

$$\text{Q-factor} = \frac{\text{Reactance of the coil at resonance}}{\text{Resistance of the circuit}} = \frac{L\omega_0}{R}$$

Q-factor determines the degree of selectivity of the circuit while tuning. This is because, for larger values of Q-factor the frequency response curve of the circuit is a steep narrow peak. For smaller values of Q-factor, the frequency response curve is quite flat (Fig. 13.18).

Sharpness of resonance of a series L-C-R circuit.

The sharpness of resonance is defined as the rapidity with which the current falls from its resonant value (E_0/R) *with change in applied freqeuency*.

A curve with low value of R falls very sharply on either side than that with high value of R (Fig. 13.18). Resonance in the first case is said to be *sharper* than in the second.

Vector Diagram. A vector diagram of a series LCR circuit is shown in Fig. 13.19. Since L-C-R are connected in series, the current through each is same. Let E_R. E_L and E_C be the potential drops across resistance, inductance and capacitance.

The vector $E_R = I_0R$ is *in phase with* the current.

The vector $E_L = \omega LI_0$ is 90° *advance of* the current.

The vector $E_C = I_0/\omega C$, is 90° *behind* the current.

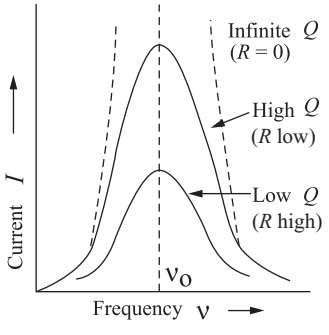


FIG. 13.18

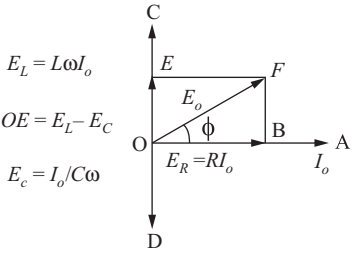


FIG. 13.19

Let the vectors *OB*, *OC* and *OD* represent E_R, E_L and E_C. If $E_L > E_C$ the resultant of these two is $(E_L - E_C)$. This is represented by *OE*.

$$OE = OC - OD$$

$$= I_0\left(L\omega - \frac{1}{C\omega}\right); \text{ where } L\omega > \frac{1}{C\omega}$$

$$E_0 = [OB^2 + BF^2]^{\frac{1}{2}} = [OB^2 + OE^2]^{\frac{1}{2}}$$

$$= \left[I_0^2R^2 + I_0^2\left(L\omega - \frac{1}{C\omega}\right)^2\right]^{\frac{1}{2}}$$

$$= I_0\left[\left(L\omega - \frac{1}{C\omega}\right)^2 + R^2\right]^{\frac{1}{2}}$$

$$I_0 = \frac{E_0}{\sqrt{\left[\left(L\omega - \frac{1}{C\omega}\right)^2 + R^2\right]}}$$

The current lags behind the applied voltage by ϕ given as

$$\phi = \tan^{-1}\left(\frac{L\omega - 1/C\omega}{R}\right)$$

j operator method

Use of operator j in study of A.C. Circuits

The operator *j* is defined as a quantity which is numerically equal to $\sqrt{(-1)}$ and which represents the rotation of a vector through 90° in anticlockwise direction, $-j$ represents the rotation of a vector through 90° in clockwise direction. The above facts about *j* are helpful in studying the A.C. circuits. We know that in A.C. circuits, E_L and E_C always lie at 90° in anticlockwise and clockwise direction respectively with respect to E_R (Fig. 13.20). Hence total emf of a circuit having *L*, *C*, *R* will be

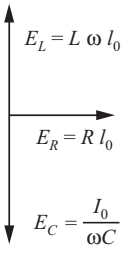


FIG. 13.20

$$E = E_R + jE_L - jE_C$$

Ordinarily a source of alternating e.m.f. *E* is denoted by E_0 sin ωt.

This is actually the imaginary part of the complex form of alternating e.m.f.,

$$E = E_0 e^{j\omega t}$$

The instantaneous current in the A.C. circuit has been expressed as $I = I_0$ sin $(\omega t - \phi)$. This expression is the imaginary part of the complex current given by

$$I = I_0\, e^{j(\omega t - \phi)}$$

Since the voltage across the inductor leads the current passing through it by 90°, the inductive reactance ω*L* can be written as $j\omega L$. $Z_L = j\omega L = jX_L$

Since the voltage across the capacitor lags the current passing through it by 90°, the capacitive reactance $1/\omega C$ can be written as

$$-j/\omega C = 1/j\omega C.\ Z_c = \frac{1}{j\omega C} = -\frac{j}{\omega C} = -jX_C$$

A complex impedance can be written as the sum of a real term and imaginary term which are to be called resistance and complex reactance respectively,

$$Z = R + jX$$

where $X = X_L - X_C$ is the effective reactance in the circuit.

LCR Circuit (Series Resonance Circuit) [Second Method]

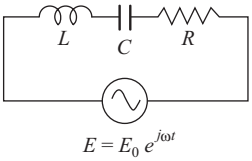


FIG. 13.21

Consider a circuit containing an inductance L, a capacitance C and a resistance R joined in series. This series circuit is connected to an AC supply given by

$$E = E_0 e^{j\omega t} \quad \text{(Fig. 13.21)} \qquad ...(1)$$

The total complex impedance is

$$Z = Z_R + Z_L + Z_C$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$= \sqrt{R^2 + (\omega L - 1/\omega C)^2}\; e^{j\phi}$$

where $\tan\phi = \dfrac{(\omega L - 1/\omega C)}{R}$ ...(2)

Using Ohm's law in complex form, the 'complex' current in the circuit is

$$I = \frac{E}{Z} = \frac{E_0 e^{j\omega t}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}\; e^{j\phi}}$$

$$\therefore \quad I = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}\, e^{j(\omega t - \phi)} \qquad ...(3)$$

But $$I_0 = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\therefore \quad I = I_0 e^{j(\omega t - \phi)} \qquad ...(4)$$

The actual emf is the imaginary part of the equivalent complex emf. Hence the actual current in the circuit is obtained by taking the imaginary part of the above 'complex' current.

$$\therefore \quad i = Im,\,(I) = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}\, \sin(\omega t - \phi) \qquad ...(5)$$

The equivalent impedance of the series LCR circuit is

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The current 'lags' behind the voltage by an angle

$$\phi = \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}$$

Example 1. *A circuit consists of a non-inductive resistance of 50 ohms, an inductance of 0.3 henry, and a resistance of 2 ohms and a capacitor of 40 micro-farad in series and is supplied with 200 volts at 50 Hz. Find the impedance, the current, lag or lead, and the power in the circuit.*

Solution. Total resistance = $R = 50 + 2 = 52\ \Omega$

$$C = 40 \times 10^{-6}\ \text{F},\ L = 0.3\ \text{H},\ \omega = 2\pi v = 2\pi \times 50 = 100\ \pi$$

$$E_{r.m.s.} = 200\ \text{V}.$$

$$\text{Impedance} = Z = \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}$$

$$= \sqrt{(52)^2 + \left(0.3 \times 100\pi - \frac{1}{40 \times 10^{-6} \times 100\pi}\right)^2}$$

$$= 53.97\ \Omega$$

$$\therefore \quad \text{current } I_{\text{r.m.s.}} = \frac{E_{rms}}{Z} = \frac{200}{53.97} = 3.71\ \text{A}.$$

$$\therefore \quad I_{\max} = 3.71 \times \sqrt{2}\ A$$

Now,
$$X_L = \omega L = (100\,\pi) \times 0.3 = 94.2\ \Omega$$

$$X_C = \frac{-1}{C\omega} = \frac{1}{(40 \times 10^{-6}) \times 100\pi} = 79.6\ \Omega$$

Since $X_L > X_C$, the current lags behind the applied e.m.f.

$$\phi = \tan^{-1}\frac{\omega L - 1/\omega C}{R} = \tan^{-1}\left(\frac{94.2 - 79.6}{52}\right) = \tan^{-1}(0.28) = 15°\ 39'$$

$$\text{Power factor } \cos\phi = \frac{R}{\sqrt{[R^2 + (\omega L - 1/\omega C)^2]}}$$

$$= \frac{52}{[(52)^2 + (14.6)^2]^{1/2}} = 0.964$$

Apparent power = $E_{rms} \times I_{rms} = 200 \times 3.71 = 742$ V.A.

True power = Apparent power × Power factor = $742 \times 0.964 = 716$ watt

Example 2. *An alternating potential of 100 volt and 50 hertz is applied across a series circuit having an inductance of 5 henry, a resistance of 100 ohm and a variable capacitance. At what value of capacitance will the current in the circuit be in phase with the applied voltage? Calculate the current in this condition. What will be the potential differences across the resistance, inductance and capacitance?*

Solution. For resonance, $\omega L = \frac{1}{\omega C}$ or $C = \frac{1}{\omega^2 L}$

Here, $\omega = 2\pi\, v = 100\ \pi$, $L = 5\ H$

$$\therefore \quad C = \frac{1}{(100\pi)^2 \times 5} = 2 \times 10^{-6}\ \text{farad} = 2\mu\text{F}$$

$$\left.\begin{matrix}\text{The current} \\ \text{at resonance}\end{matrix}\right\} = I_{rms} = \frac{E_{rms}}{R} = \frac{100}{100} = 1.0\ A$$

P.D. across $R = I_{rms} \times R = 1.0 \times 100 =$ **100 volt (rms)**,

P.D. across $L = I_{rms} \times \omega L = 1.0 \times (100\pi) \times 5 =$ **1570 volt (rms)**

P.D. across $C = I_{rms} \times \dfrac{1}{C\omega} = 1.0 \times \dfrac{1}{(2 \times 10^{-6}) \times 100\pi}$

= **1570 volt (rms)**.

The voltages across the inductor and capacitor are much greater than the applied voltage. But they differ in phase by 180°. So their algebraic sum is zero.

13.4 PARALLEL RESONANT CIRCUIT

Here, capacitor C is connected in parallel to the series combination of resistance R and inductance L. The combination is connected across the AC source (Fig. 13.22). The applied voltage is sinusoidal, represented by

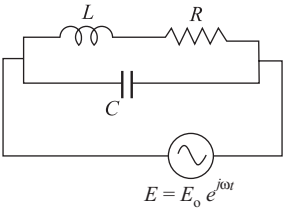


FIG. 13.22

$$E = E_0 e^{j\omega t}$$

Complex impedance of L - branch

$$Z_1 = \mathrm{R} + jL\omega$$

Complex impedance of C- branch

$$Z_2 = \frac{1}{jC\omega}$$

Z_1 and Z_2 are in parallel.

$$\frac{1}{Z} = \frac{1}{R + j\omega L} + \frac{1}{1/j\omega C} = \frac{1}{R + j\omega L} + j\omega C$$

$$= \frac{R - j\omega L}{(R + j\omega L) \times (R - j\omega L)} + j\omega C$$

$$= \frac{R}{R^2 + (L\omega)^2} + j\left[C\omega - \frac{L\omega}{R^2 + (L\omega)^2}\right]$$

The current $I = E/Z = E \times \dfrac{1}{Z}$

$\therefore$ $$I = E\left[\frac{R}{R^2 + (L\omega)^2} + j\left(C\omega - \frac{L\omega}{R^2 + (L\omega)^2}\right)\right]$$

Let $A \cos\phi = \dfrac{R}{R^2 + (L\omega)^2}$; $A \sin\phi = C\omega - \dfrac{L\omega}{R^2 + (L\omega)^2}$

$\therefore$ $$I = E\,(A\cos\phi + j\,A\sin\phi) = EAe^{j\phi} = E_0\,A\,e^{j(\omega t + \phi)}$$

where $$\phi = \tan^{-1}\frac{C\omega - \left(\dfrac{L\omega}{R^2 + (L\omega)^2}\right)}{\left(\dfrac{R}{R^2 + (L\omega)^2}\right)};$$

$$A^2 = \frac{R^2}{(R^2 + \omega^2 L^2)^2} + \left(C\omega - \frac{L\omega}{R^2 + \omega^2 L^2} \right)^2$$

The magnitude of the admittance

$$Y = \frac{1}{Z} = \frac{\sqrt{[R^2 + (\omega CR^2 + \omega^3 L^2\, C - \omega L)^2]}}{R^2 + \omega^2 L^2}$$

The admittance will be minimum, when

$$\omega CR^2 + \omega^3 L^2 C - \omega L = 0$$

or $$\omega = \omega_0 = \sqrt{\left[\frac{1}{LC} - \frac{R^2}{L^2} \right]}$$

or $$v_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

This is the resonant frequency of the circuit.

If R is very small so that $\frac{R^2}{L^2}$ is negligible compared to $\frac{1}{LC}$,

$$v_0 = \frac{1}{2\pi \sqrt{(LC)}}$$

At such a minimum admittance, *i.e.*, maximum impedance, the circuit current is minimum.

The graph between current and frequency is shown in Fig. 13.23.

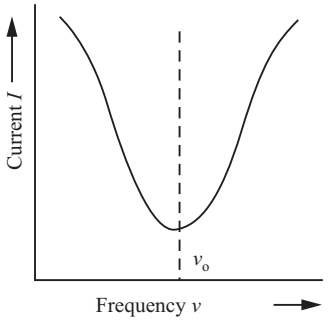


FIG. 13.23

Impedance at Resonance

At resonance, $$Z = \frac{R^2 + (L\omega)^2}{R}$$

But $$R^2 + (L\omega)^2 = \frac{L}{C} \text{ at resonance}$$

$\therefore$ $$Z = \frac{L}{RC}$$

Thus smaller the resistance R, larger is the impedance. If R is negligible, the mpedance is infinite at resonance.

Rejector Circuit. The parallel resonant circuit does not allow the current of the same frequency as the natural frequency of the circuit. Thus it can be used to suppress the current of this particular frequency out of currents of many other frequencies. Hence the circuit is known as a 'rejector' or 'filter' circuit.

Comparison between series and parallel resonant circuits

The behaviour of a parallel resonant circuit is strikingly different from that of a series resonant circuit. In both cases the impedance is resistive but, whereas parallel resonance implies maximum impedance, series resonance implies minimum impedance.

Series resosnant circuit	*Parallel resosnant circuit*
1. An acceptor circuit	A rejector circuit
2. Resosnant frequency $v_r = \frac{1}{2\pi\sqrt{LC}}$	Resonant frequency $v_r = \frac{1}{2\pi\sqrt{LC}}$
3. At resonance the impedance is a minimum equal to the resistance in the circuit.	At resonance the impedance is maximum nearly equal to infinity
4. Selective	Selective
5. Used in the tuning circuit to separate the wanted frequency from the incoming frequencies by offering low impedance at that frequency.	Used to present a maximum impedance to the wanted frequency, usually in the plate circuit of valves.

Example 1. *A coil of self inductance 2 milli-henry and resistance 15 ohm is connected in parallel with a capacitance of 0.001* μ*F. Find* (*a*) *the frequency at which the current from an a.c. source to this circuit is minimum*, (*b*) *the peak-value of this make-up current if the peak value of supply voltage is 2 volt.*

Solution. (*a*) Current is minimum at resonant frequency.

$$v_0 = \frac{1}{2\pi}\left(\frac{1}{LC} - \frac{R^2}{L^2}\right)^{1/2} = \frac{1}{2\pi}\left[\frac{1}{(2\times10^{-3})\,(0.001\times10^{-6})} - \frac{(15)^2}{(2\times10^{-3})^2}\right]^{1/2} = 112590 \text{ Hz}$$

Aliter.

$$v_0 = \frac{1}{2\pi}\frac{1}{\sqrt{(LC)}}\left(\text{as } \frac{R^2}{L^2} << \frac{1}{LC}\right) = \frac{1}{2\times3.14\times\sqrt{(2\times10^{-3}\times0.001\times10^{-6})}} = 112596 \text{ Hz}$$

(*b*) Peak-value of make-up current at resonance

$$= \frac{\text{peak value of applied e.m.f.}}{\text{impedance at resonance}} = \frac{E_0}{L/CR} = \frac{E_0 CR}{L}$$

$$= \frac{2\times(0.001\times10^{-6})\times15}{2\times10^{-3}} = 15\times10^{-6}\,A$$

13.5 POWER IN AC CIRCUIT CONTAINING RESISTANCE, INDUCTANCE AND CAPACITANCE

Consider an ac circuit containing resistance, inductance and capacitance. E and I vary continuously with time. Therefore power is calculated at any instant and then its mean is calculated over a complete cycle.

The instantaneous values of the voltage and current are given by

$$E = E_0 \sin \omega t,$$

$$I = I_0 \sin (\omega t - \phi)$$

where ϕ is the phase difference between current and voltage.

Hence power at any instant is

$$E \times I = E_0 I_0 \sin \omega t \sin (\omega t - \phi)$$

$$= \frac{1}{2} E_0 I_0 [\cos \phi - \cos (2\omega t - \phi)] \quad ...(1)$$

Average power consumed over one complete cycle is

$$P = \frac{\int_0^T E\ I\ dt}{\int_0^T dt}$$

$$= \frac{\int_0^T \frac{1}{2} E_0\ I_0 [\cos \phi - \cos (2\omega t - \phi)]\, dt}{T}$$

$$= \frac{1}{2} \frac{E_0 I_0}{T} \left[(\cos \phi)\, t - \frac{\sin (2\omega t - \phi)}{2\omega} \right]_0^T$$

$$= \frac{1}{2} \frac{E_0 I_0}{T} \left[(\cos \phi)\, T - 0 - \frac{\sin (2\omega T - \phi)}{2\omega} + \frac{\sin (-\phi)}{2\omega} \right]$$

Now, $T = \frac{2\pi}{\omega}$ and $\sin (4\pi - \phi) = \sin (-\phi)$

$$P = \frac{1}{2} \frac{E_0 I_0 \omega}{2\pi} \left[(\cos \phi) \frac{2\pi}{\omega} - \frac{\sin (-\phi)}{2\omega} + \frac{\sin (-\phi)}{2\omega} \right]$$

$$= \frac{1}{2} E_0 I_0 \cos \phi$$

$$= \frac{E_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos \phi$$

$$= E_{r.m.s.}\ I_{r.m.s.} \cdot \cos \phi \quad \ldots(2)$$

average power = (virtual volts) × (virtual amperes) × cos ϕ

The term (*Virtual volts* × *Virtual amperes*) *is called* apparent power *and cos* ϕ *is called the power factor. Thus*

True power = apparent power × power factor.

Evidently, the *power factor is the ratio of the true power to the apparent power.*

As cos ϕ is the factor by which the product of the r.m.s. values of the voltage and current must be multiplied to give the power dissipated, it is known as the 'power factor' of the circuit. For a circuit containing resistance, capacitance and inductance in series,

$$\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R}$$

From Fig. 13.24, the expression for the power factor is

$$\cos \phi = \frac{R}{\sqrt{\left\{ R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right\}}}$$

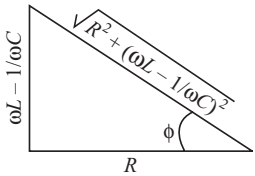


FIG. 13.24

Special cases.

(1) In a purely *resistive circuit*, ϕ = 0 or cos ϕ = 1,

∴ true power = $E_v \times I_v$

(2) In a *purely inductive circuit*, current lags behind the applied emf by 90° so that $\phi = 90°$ or $\cos \phi = 0$.

Thus true power consumed = 0

(3) In a *purely capacitive circuit*, current leads the applied voltage by 90° so that $\phi = -90°$ or $\cos (-90°) = \cos 90° = 0$

$\therefore$ true power = 0

(4) In an ac circuit containing a resistance and inductance in series.

$$\textit{Power factor}, \cos \phi = \frac{R}{\sqrt{R^2 + (L\omega)^2}}$$

(5) In an ac circuit containing a capacitance C and a resistance R in series,

$$\cos \phi = \frac{R}{\sqrt{\left(\frac{1}{C^2\omega^2} + R^2\right)}}$$

Wattless current

The average power dissipated during a complete cycle is $E_v I_v \cdot \cos \phi$

The current in A.C. circuit is said to be wattless when the average power consumed in the circuit is zero.

If an ac circuit is purely inductive or purely capacitive with no ohmic resistance, phase angle $\phi = \pi/2$ so that $\cos \phi = 0$ or the power consumed is zero. The current in such a circuit does not perform any useful work and is rightly called the *wattless* or *idle current*. In this situation, the circuit does not consume any power, though it offers a resistance to the flow of alternating current in it. It is the principle of choke coil.

Example 1. *An alternating voltage of 10 volts at 100 Hz is applied to a choke of inductance 5 henry and of resistance 200 ohms. Find the power factor of the coil and the power absorbed.*

Solution. Here, $E_v = 10\ V$, $v = 100\ Hz$, $L = 5\ H$, and $R = 200\ \Omega$

$$\text{Power factor}, \cos \phi = \frac{R}{\sqrt{R^2 + \omega^2 L^2}} = \frac{200}{\sqrt{(200)^2 + (2\pi \times 100 \times 5)^2}} = 0.062$$

$$\text{Power absorbed} = E_v \cdot I_v \cdot \cos \phi = E_v \cdot \frac{E_v}{Z} \cos \phi$$

$$= \frac{(E_v)^2}{\sqrt{R^2 + \omega^2 L^2}} \cos \phi = \frac{10 \times 10\ (0.062)}{\sqrt{(200)^2 + (2\pi \times 100 \times 5)^2}} = 0.00189 \text{ W}$$

Example 2. *A source of e.m.f. 50 V, r.m.s. is connected across an air cored coil. When the supply frequency is 50 Hz the power consumed is found to be 100 W, whereas when the frequency is increased to 100 Hz, the power becomes 50 W. Find the self-inductance and the resistance of the coil.*

Sol. Let the inductance and resistance required be L and R respectively.

$$I_v = \frac{E_v}{\sqrt{(R^2 + \omega^2 L^2)}}$$

When the supply frequency is 50 Hz, $\omega = 2\pi \times 50 = 100\ \pi$

$$I_v' = \frac{50}{\sqrt{R^2 + (100\pi)^2 L^2}}$$